

Section 2.3 Evaluating Limits

How much is $\frac{1}{0}$?

Ex ① Let $f(x) = \frac{1}{x^2}$. Notice that $f(0)$ is undefined.
What happens to $f(x)$ if x is close to zero?

$$x = 0.1 \rightarrow f(0.1) = \frac{1}{0.1^2} = 100$$

$$x = 0.01 \rightarrow f(0.01) = \frac{1}{0.01^2} = 10,000$$

$$x = -0.01 \rightarrow f(-0.01) = \frac{1}{(-0.01)^2} = 10,000$$

As x gets close to zero, $f(x)$ is getting larger.

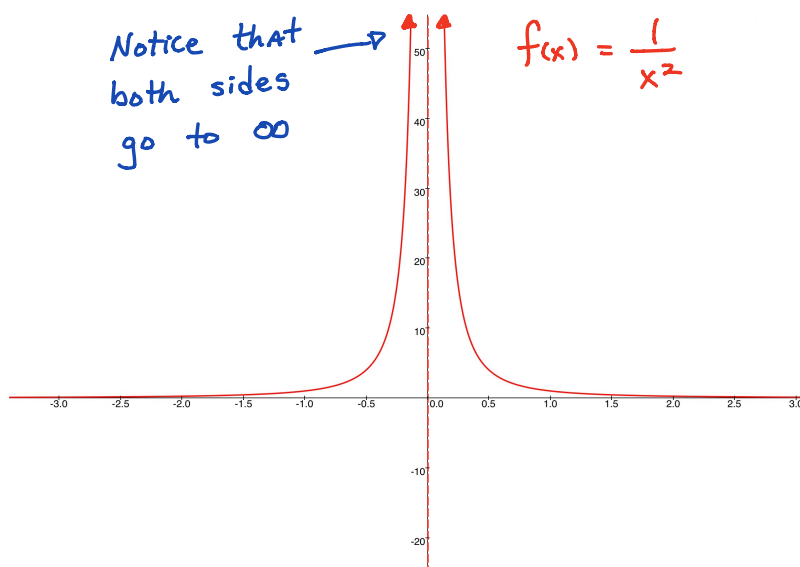
$\downarrow$ $\downarrow$

$x \rightarrow 0$ $f(x) \rightarrow \infty$

or The limit of $f(x)$ as x goes to zero is infinity.

Notation $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$

Note $x \rightarrow a$ means that x approaches a from both sides of a .



To evaluate $\lim_{x \rightarrow a} f(x)$:

Try to evaluate $f(a)$. If you get a real number, then that is the limit.

If $f(a)$ has division by zero, use algebra to simplify the function and then substitute $x=a$ again.

The graph of $f(x)$ can help verify an answer, but does not actually prove anything.

$$\underline{\text{Ex}} \text{ ② } \lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} \rightarrow \frac{4^2 - 16}{4 - 4} = \frac{0}{0} \text{ So, do Algebra.}$$

$$= \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{x-4}$$

$$= \lim_{x \rightarrow 4} (x+4) = 4+4 = \textcircled{8}$$

$$\underline{\text{Ex}} \text{ ③ } \lim_{x \rightarrow 3} \frac{x-3}{x^2-9} \rightarrow \frac{3-3}{3^2-9} = \frac{0}{0} \text{ So, do Algebra.}$$

$$= \lim_{x \rightarrow 3} \frac{x-3}{(x-3)(x+3)}$$

$$= \lim_{x \rightarrow 3} \frac{1}{x+3} = \frac{1}{3+3} = \textcircled{\frac{1}{6}}$$

$$\underline{\text{Ex}} \text{ ④ } \lim_{x \rightarrow 100} \frac{\sqrt{x} - 10}{x - 100} \text{ use the conjugate } \frac{\sqrt{x} + 10}{\sqrt{x} + 10}$$

$$\lim_{x \rightarrow 100} \frac{(\sqrt{x} - 10)(\sqrt{x} + 10)}{(x - 100)(\sqrt{x} + 10)}$$

$$= \lim_{x \rightarrow 100} \frac{x - 100}{(x - 100)(\sqrt{x} + 10)}$$

$$= \lim_{x \rightarrow 100} \frac{1}{\sqrt{x} + 10} = \frac{1}{\sqrt{100} + 10} = \textcircled{\frac{1}{20}}$$

Ex ⑤ $\lim_{h \rightarrow 0} \frac{(5+h)^2 - 5^2}{h}$ multiply this out

$$= \lim_{h \rightarrow 0} \frac{25 + 10h + h^2 - 25}{h}$$

$$= \lim_{h \rightarrow 0} \frac{10h + h^2}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(10+h)}{h} = \lim_{h \rightarrow 0} (10+h) = \textcircled{10}$$

Ex ⑥ $\lim_{x \rightarrow -7} \frac{\frac{1}{x} + \frac{1}{7}}{7+x}$ use LCD = 7x

$$\lim_{x \rightarrow -7} \frac{\left(\frac{1}{x} + \frac{1}{7}\right) 7x}{(7+x) 7x}$$

$$= \lim_{x \rightarrow -7} \frac{7+x}{(7+x) \cdot 7x}$$

$$= \lim_{x \rightarrow -7} \frac{1}{7x} = \textcircled{-\frac{1}{49}}$$

Ex ⑦ $\lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$

$$= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} \quad \text{cancel } h \text{ from each term}$$

$$= \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2$$

$$= 3x^2 + 3x(0) + 0^2 = \textcircled{3x^2}$$

$$\underline{\text{Ex}} \textcircled{8} \quad \lim_{x \rightarrow -2} \frac{x^3 + 8}{x + 2} \quad \leftarrow A^3 + B^3 = (A+B)(A^2 - AB + B^2)$$

$$= \lim_{x \rightarrow -2} \frac{(x+2)(x^2 - 2x + 4)}{x+2}$$

$$= \lim_{x \rightarrow -2} x^2 - 2x + 4$$

$$= (-2)^2 - 2(-2) + 4$$

$$= 4 + 4 + 4 = \textcircled{12}$$

$$\underline{\text{Ex}} \textcircled{9} \quad \lim_{h \rightarrow 0} \frac{\frac{1}{(x+h)^2} - \frac{1}{x^2}}{h} \quad \text{LCD} = x^2(x+h)^2$$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{1}{(x+h)^2} - \frac{1}{x^2} \right)}{(h)} \cdot \frac{x^2(x+h)^2}{x^2(x+h)^2}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 - (x+h)^2}{h \cdot x^2 (x+h)^2} \quad \leftarrow \text{multiply it out}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 - x^2 - 2xh - h^2}{h \cdot x^2 (x+h)^2}$$

$$= \lim_{h \rightarrow 0} \frac{-2xh - h^2}{h \cdot x^2 (x+h)^2} \quad \text{cancel } h \text{ from each term}$$

$$= \lim_{h \rightarrow 0} \frac{-2x - h}{x^2 (x+h)^2}$$

$$= \frac{-2x - 0}{x^2 (x+0)^2}$$

$$= \frac{-2x}{x^4} = \frac{-2}{x^3}$$